

Arturo Vaine

AI Engineer | Full Stack Developer

AI Engineer and Full Stack Developer with 6+ years shipping production LLM agents, RAG pipelines, and data infrastructure for clients across the US, Europe, and Brazil.

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SKILLS

AI/ML: LLMs, RAG, MCP, LangChain, LangGraph, Embeddings, Vector DBs (pgvector, MongoDB Atlas), ETL Pipelines, GenAI
Programming: Python, JavaScript, TypeScript, Node.js
Frameworks: React, Express.js, FastAPI, Flask, Django, SQLAlchemy
Databases: MySQL, PostgreSQL, MongoDB, Firebase, BigQuery
Cloud: AWS (EC2, S3, Lambda, IAM, CloudFront), GCP (Cloud Storage, BigQuery, Cloud SQL)
DevOps: Docker, Airflow, Grafana, Jenkins, CI/CD
Tools: n8n, Shopify, Zapier, PowerBI, Metabase, Streamlit, Retool

PROFESSIONAL EXPERIENCE

Software Engineer Apr 2025 - Present
Instituto Meros do Brasil - Curitiba-PR-Brazil; Remote (freelancer)
Non-profit marine conservation program (60+ staff across 9 Brazilian states); subsidized by PETROBRAS S.A.

- Built a back-office platform (MERN) with MCP for task management and reporting across a 60+ person team
- Deployed ETL pipelines and a RAG system consolidating 2,000+ reports into a searchable knowledge base
- Automated Google Workspace processes (monthly reports, email triage) integrated with Google Cloud Platform

Full Stack Engineer Aug 2026 - Sep 2026
Colorado Industrial Packaging - Denver, CO, United States; Remote (contract)
Ad-hoc consultant; full-stack web application for the quotation process

- Migrated spreadsheet-based and legacy workflows into a maintainable web application
- Translated existing business rules and operational processes into application logic

AI Engineer May 2026 - Aug 2026
Main Street AI Advisors - California, United States; Remote (contract)

- Founding Engineer: set up the company's GitHub org and bootstrapped 7 projects; AI strategy for SMBs (CO, LA, MT, TX)
- Built an AI engine for construction projects, analyzing PDFs and images to identify bid risks
- Delivered an AI lead-qualification agent integrated with CRM at under \$20/mo in infrastructure cost
- Developed competitor-pricing tooling (Apify) and a C-level ops assistant for inbox triage and daily reporting

AI Engineer May 2025 - Nov 2025
NILG.AI - Porto, Portugal; Remote (consultant/contractor)
Tailored AI solutions to grow your business, scale faster, and cut costs, from strategy to deployment.

- Built MCP and audiovisual pipelines using Python, Zapier, webhooks, and FFmpeg to process TV episodes
- Achieved 80% time savings by automating docs processing with ETLs and vision-language models
- Developed RAG agents and chatbots (front-end React) for multiple industries, including tourism and healthcare

Data Engineer Jun 2025 - Oct 2025
LitBoat | Stealth Startup - Miami, United States; Remote (freelancer)

- Deployment of Medallion Architecture in BigQuery and APIs (SQLAlchemy, FastAPI) for data ingestion
- Integration of data pipelines and Analytics, GCP (Buckets, Postgres, BigQuery), Retool and Metabase
- Built dataviz on Metabase delivering strategic KPIs for thousands of clients (restaurants)

Full Stack Developer Sep 2024 - Feb 2025
Rise Layers | 88 Delivery Platform - Curitiba-PR-Brazil; On-site
Developer at Delivery branch (early stage startup) - Restaurant Delivery Platform (300+ restaurants)

- Built digital menu from scratch (React), to open new revenue channel for more than 300 customers (restaurants)
- Developed integration and applications: portfolio management, broadcast messages, etc. (NoSQL, Firebase)
- Deployed ETLs for: CRM sync, portfolio enrichment, data analysis - RFM matrix, ABC curve, etc. (Python)

Full Stack Developer Mar 2023 - Sep 2024
Supercampo S/A - Curitiba-PR-Brazil; On-site

- Delivered data through ETL / BI of R\$ 160MM+ annual GMV and other KPIs (MySQL, PowerBI, Streamlit)
- Supported technology innovation project (Tecnoparque), reducing company's service tax (ISS) from 5% to 2%
- Developed MERN stack for Product Information Manager (PIM) of portfolio enrichment w/ GenAI (180k+ SKUs)

Back-end Developer Oct 2022 - Mar 2023
BRG Brasil Geradores - Anápolis-GO-Brazil; Remote

- Delivered key features to enhance APIs / back-end of National Power Plant Monitoring System
- Refactored and modernized legacy Express.js codebase by transitioning to a multi-layer architecture REST API
- Documented APIs using OpenAPI notation, Swagger and markdown docs

Software Developer Jul 2019 - Dec 2021
SIMEPAR - Sistema de Tecnologia e Monitoramento Ambiental do Paraná - Curitiba-PR-Brazil; On-site
Dam Monitoring Project - overall output delivered opportunity of collecting 7-digit fees

- Developed data collection solution of technical form with ~350 inputs (HTML, Vanilla JavaScript, CSS; SQLite)
- Delivered data analysis process to check inconsistencies and generate ~900 reports (Python-Pandas)
- Designed PowerBI decision tool for monitoring KPIs: team's productivity, deliverables and scheduling

EDUCATION

Mechanical Engineering, Bachelor Degree 2016
Federal University of Technology – Paraná, UTFPR, Curitiba-PR-Brazil
Mechanical Engineering (Non-degree, visiting student; 2 semesters, ~1000h) 2014
National Institute of Applied Sciences of Lyon (INSA), Villeurbanne-France. 2013 cohort

LANGUAGES

English: Advanced **French:** Advanced

AWARDS & RECOGNITION

2021 Fellow of Leaders in Innovation Programme (LIF-7th cohort), Royal Academy of Engineering (London-UK)
2019 2nd place Global Legal Hackathon (Nous Team) - National Law Organization (OAB-PR-Brazil)
2018 1st place Ozires Silva Award for Sustainable Entrepreneurship (11th edition) - ISAE/FGV
2017 3rd place Perfumathon: Hackathon of Perfumery (Epic Team) - O Boticário
2013 1st place Scholarship selection UTFPR 2013 cohort, CAPES/BRAFITEC, Brazil-France